

DynamicFeeMock Hook Specification

Hook Bazaar Formal Specification Format

Version 1.0 — Category: FEE_MANAGEMENT — License: MIT

1. Hook State Variables (H)

Symbol	Type	Description	Initial
ϕ_{base}	uint24	Base fee (basis points)	3000
ϕ_{min}	uint24	Minimum fee bound	100
ϕ_{max}	uint24	Maximum fee bound	10000
α	uint256	Volatility coefficient	10^{16}
V_{acc}	uint256	Accumulated volatility index	0
t_{last}	uint256	Last update timestamp	0

2. Pool State Dependencies (from $\mathcal{S}_T$)

Reads: $\sqrt{P}$ (sqrtPriceX96), t_c (tick), ϕ_{lp} (lpFee), L_{act} (liquidity)

Writes: ϕ_{lp} (lpFee via `beforeSwap` return)

3. State Transition: `beforeSwap`

Preconditions:

- $L_{act} > 0$ (pool has active liquidity)
- $\phi_{min} \leq \phi_{base} \leq \phi_{max}$

Transition Equations:

$$\Delta P = \left| \sqrt{P}_{current} - \sqrt{P}_{last} \right| \quad (1)$$

$$V'_{acc} = V_{acc} + \frac{\Delta P \cdot \alpha}{10^{18}} \quad (2)$$

$$\sigma = \frac{V'_{acc}}{t_{current} - t_{last} + 1} \quad (3)$$

$$\phi_{dynamic} = \min(\phi_{max}, \max(\phi_{min}, \phi_{base} + \sigma)) \quad (4)$$

Postconditions:

- $\phi_{min} \leq \phi_{dynamic} \leq \phi_{max}$ (fee bounds invariant)
- $V'_{acc} \geq V_{acc}$ (volatility monotonic)

Return: (`selector`, $\Delta_0 = 0$, $\phi_{dynamic}$)

4. Invariants

ID	Expression	Severity
INV-1	$\phi_{min} \leq \phi_{lp} \leq \phi_{max}$	Critical
INV-2	$V'_{acc} \geq V_{acc}$	Warning
INV-3	$\phi_{base} \in [100, 10000]$	Critical

5. Constant Product Reference

From the dual-index state-space model, the tick-wise invariant holds:

$$K - X' \leq K' \leq K \quad \text{where} \quad K = X \cdot Y$$

The hook does **not** modify K ; it only adjusts ϕ_{lp} based on volatility.